

Block 13

Informed Consent for Online Survey

Study Title: Explanations for Self-Driving Cars.

Purpose of the Study: You are invited to participate in a research study examining the ability of explanations in self-driving cars. You must be at least 18 years of age to participate.

Procedures: If you choose to participate, you will answer some questions after watching short videos pertaining to the actions of a self-driving car. The study is expected to take approximately 12 minutes.

Potential Risks and Benefits: Your participation in the survey will not expose you to harmful, disturbing, or offensive content. There are no specific benefits or risks associated with participating in the survey.

Confidentiality: The data collected in this study is completely anonymous. No personally identifiable information will be collected and the information you

choose to provide in this study cannot be connected back to you. Results from this study may be used for any purposes by the survey owner including, to publish or present academic papers. All data collected will be owned by the survey owner. The videos shown during the survey are for testing and experimental purposes only. The videos, and any other information, shown to you during the survey remain the sole and exclusive intellectual property of the survey owner. You are hereby prohibited from using the videos for any purpose other than responding to the survey.

Voluntary Participation: Your participation in this study is voluntary and you may choose to not participate or end your participation at any time without penalty.

Consent: I have read and understand the above consent form. I certify that I am 18 years old or older. By clicking the “Next” button to enter the survey, I indicate my willingness to voluntarily take part in this study.

Preamble

Please do not take this study on a mobile phone, the videos and text will not display correctly

You are free to leave the study anytime, but if completing the study please do so in one sitting.

The survey will take around 12 minutes depending on your speed.

Thank you for your participation!

Name

Please give your Prolific ID

Instructions

Instructions

You will be shown 3 scenarios of **an** **experimental** autonomous vehicle (i.e., AV) being tested

(along with 1 practice question).

The AV is a self-driving car, you should not expect it to have the same performance as state-of-the-art AVs.

The videos will be from the point of view of the driver's seat in the car.

Your task is to:

Step 1: Carefully view the video and imagine yourself in the driver's position.

Step 2: Select, from two choices, the reason for the behaviour of the AV which best fits your current beliefs. Then, rate how confident you are in this answer.

Step 3: Answer a question predicting what the AV would do in an alternative situation. Then, give a confidence score for this prediction, and a free-form text rationale explaining your prediction.

-- click to next page --

Step 4: View the same video, ***but this time with an explanation*** of what concepts the AV is detecting.

Step 5: In light of the explanation, repeat **Step 2**.

Step 6: In light of the explanation, repeat **Step 3**.

Most of the videos do not have great quality, but please try your best.

Instructions continued

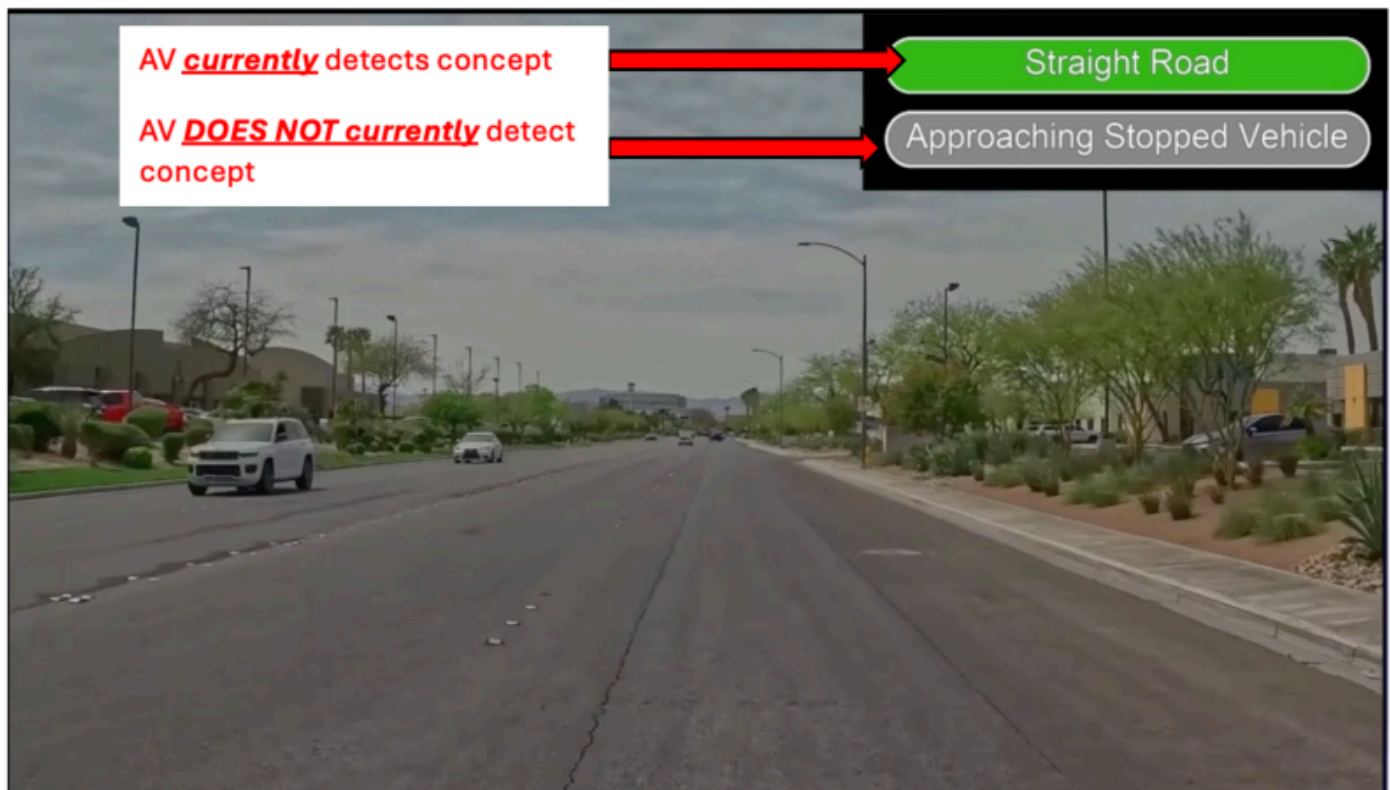
Instructions for interpreting explanations

When you see the video a second time it will contain **real-time explanations** of the AV behavior. Specifically, you will see **concepts** that it is detecting:

- If a concept **is currently detected**, you will see it **highlighted in green**.
- If a concept **is NOT currently detected**, it will be **gray in color**.
- You will only see the **two most relevant concepts per video**.
- The concepts may turn green or gray at anytime during the video.

Important: It is possible a concept may never activate.

For Example see this image below



Final instructions

Final (important) Instructions

Each scenario you see is with a **different automated vehicle (AV)**, you should consider them all separately and ***DO NOT conflate the ability of one AV to the next AV.***

For example, if you think one video demonstrates the AV is very good (or bad) at detecting a certain concept, you

should not assume the AV in the next video is the same.

Again, this is an **experimental AV**, do not have expectations of its abilities, it may correctly or incorrectly detect concepts at any time.

Transition to practice question

Now, please practice on the next question

Practice question

Step 1:

Please watch this video which shows the driver's point of view in a self-driving car.

practice

Eoin



Watch on

Step 2:

Which of the following best fit your current beliefs about this AV?

- ☐ The AV was driving straight partly because it detected a straight road.
- ☐ The AV was driving straight because it was not detecting any other vehicles ahead.

Provide a confidence rating for this.

	Not confident at all			Neutral			Extremely confident
How confident are you in this choice?	☐	☐	☐	☐	☐	☐	☐

Step 3:

If there was a traffic jam ahead blocking this AV's path, do you think it would have kept moving forward?

- ☐ Yes
- ☐ No

Provide a confidence rating for this prediction.

	Not confident at all			Neutral			Extremely confident
How confident are you in this prediction?	☐	☐	☐	☐	☐	☐	☐

In your own words, and in as much detail as possible, please explain your reasoning for this prediction in **Step 3**.



Step 4: (video with explanation)

Please watch this video which shows the driver's point of view in a self-driving car, **now with the concepts it currently detects**.



Watch on

Step 5:

After seeing the explanation, which of the following best fit your current beliefs about this AV?

- ☐ The AV was driving straight partly because it detected a straight road.
- ☐ The AV was driving straight because it was not detecting any other vehicles ahead.

Now provide a confidence rating for this.

How confident are you in this prediction?

	Not confident at all			Neutral			Extremely confident
	☐	☐	☐	☐	☐	☐	☐

Step 6:

After seeing the explanation, if there was a traffic jam ahead blocking this AV's path, do you think it would have kept moving forward?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 6**.

Start study

Please note **you are being paid above the average rate on Prolific** because although this study is short, it requires a high degree of concentration and effort, please only proceed if you are confident you understood the instructions.

In particular, please take care with the free form text responses you give, they are important to understand your

reasoning.

start study preamble

Click Next to Begin the Real Survey

CLOSE

Step 1:

Please watch this video which shows the driver's point of view in a self-driving car.

Note there is a **pedestrian pickup drop-off zone** at the end of the video ahead to the left which the car can detect.

CS

userstudytemp



Watch on

Step 2:

Which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped because of the pedestrian pickup drop-off zone just ahead.

- ☐ The AV stopped because of the parked cars close by to the right.

Now provide a confidence rating for this prediction.

	Not confident at all			Neutral			Extremely confident
How confident are you in this prediction?	☐	☐	☐	☐	☐	☐	☐

Step 3:

If this AV was slightly to the left, do you think it still would have kept moving forwards?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 3**.

Step 4: (video with explanation)

Please watch this video which shows the driver's point of view in a self-driving car, **now with the concepts it currently detects**.

Note there is a **pedestrian pickup drop-off zone** at the end of the video ahead to the left which the car can detect.

CSX

userstudytemp



Watch on

Step 5:

After seeing the explanation, which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped because of the pedestrian pickup drop-off zone just ahead.
- ☐ The AV stopped because of the parked cars close by to the right.

Now provide a confidence rating for this prediction.

How confident are you in this prediction?

	Not confident at all			Neutral			Extremely confident
	☐	☐	☐	☐	☐	☐	☐

Step 6:

After seeing the explanation, if this AV was slightly to the left, do you think it still would have kept moving forwards?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 6**.

ASV

Step 1:

Please watch this video which shows the driver's point of view in a self-driving car.

There are **two clips** in this video for added context on this AV.

ASV base

TallWaters



Watch on

Step 2:

Which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped in the second clip to avoid colliding with the traffic cone.
- ☐ The AV stopping in the second clip had nothing to do with the cone, it randomly hallucinated a stopped vehicle ahead.

Now provide a confidence rating for this prediction.

How confident are you in this prediction?

	Not confident at all			Neutral			Extremely confident
	☐	☐	☐	☐	☐	☐	☐

Step 3:

If the traffic cone was not in front of this AV in the second clip, do you think this AV would have kept moving forwards?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 3**.

Step 4: (video with explanation)

Please watch this video which shows the driver's point of view in a self-driving car, **now with the concepts it currently detects**.

There are **two clips** in this video for added context on this AV.

ASV XAI

TallWaters



Watch on

Step 5:

After seeing the explanation, which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped in the second clip to avoid colliding with the traffic cone.
- ☐ The AV stopping in the second clip had nothing to do with the cone, it randomly hallucinated a stopped vehicle ahead.

Now provide a confidence rating for this prediction.

How confident are you in this prediction?

	Not confident at all			Neutral			Extremely confident
	☐	☐	☐	☐	☐	☐	☐

Step 6:

After seeing the explanation, if the traffic cone was not in front of this AV in the second clip, do you think this AV would have kept moving forwards?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 6**.

BIKE

Step 1:

Please watch this video which shows the driver's point of view in a self-driving car.

bike base
TallWaters



Watch on

Step 2:

Which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped because it detected the cyclist.
- ☐ The AV stopped because of its automatic emergency braking system.

Now provide a confidence rating for this decision.

	Not confident at all			Neutral			Extremely confident
How confident are you in this choice?	☐	☐	☐	☐	☐	☐	☐

Step 3:

Based on on this interaction, do you trust that this AV will reliably slow down for cyclists in similar situations in the future?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 3**.

Step 4: (video with explanation)

Please watch this video which shows the driver's point of view in a self-driving car, **now with the concepts it currently detects**.

bike xai

TallWaters



Watch on

Step 5:

After seeing the explanation, which of the following best fit your current beliefs about this AV?

- ☐ The AV stopped because it detected the cyclist.
- ☐ The AV stopped because of the automatic emergency braking system.

Now provide a confidence rating for this decision.

How confident are you in this prediction?

	Not confident at all			Neutral			Extremely confident
	☐	☐	☐	☐	☐	☐	☐

Step 6:

After seeing the explanation, based on on this interaction, do you trust that this AV will reliably slow down for cyclists in similar situations in the future?

- ☐ Yes
- ☐ No

Now provide a confidence rating for this prediction.

Not
confident
at all

Neutral

Extremely
confident

How confident are
you in this
prediction?



In your own words, and in as much detail as possible,
please explain your reasoning for this prediction in **Step 6**.

End

**Click next to end the
survey (thank you!)**

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